A Recursive H-Representation for Separable Choice

Online appendix to *Entangled vs. Separable Choice*

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Abstract

We turn the replica characterization of separable choice in [2] into a finite system of linear inequalities in observed joint choice probabilities. For any finite domain with two decision makers and unrestricted feasible local plans, a recursion gives every coefficient and an accounting certificate for a violated inequality. A binary example reduces the recursion to intersecting intervals for a common population share. We also give a finite formula for evaluating a fixed score when a new menu is added, and a separate parametric characterization for the full three-alternative menu domain.

1 Separable choice

There are two decision makers, indexed by $t = 1, 2$. Each has a finite nonempty set X_t of local covariate values. At $x_t \in X_t$, the feasible actions form a finite nonempty menu $M_t(x_t)$. A choice function $c^t \in \mathcal{C}^t$ chooses one feasible action at every own covariate. We impose no preference maximization, WARP, transitivity, or restrictions connecting these choices. Write $\mathcal{D} = (X_t, M_t)_{t=1,2}$ for the domain and let $\mathcal{C}^t$ include every feasible choice function.

The observation $\rho(y_1, y_2; x_1, x_2)$ is a joint probability distribution at every covariate pair. Every block is nonnegative and normalized. Separability implies no signalling: holding a person's own covariate fixed, her marginal choice distribution is invariant to the other person's covariate. Every pair in $X_1 \times X_2$ is observed.

Definition 1 (Separability)**.** The rule ρ is separable if there is one probability distribution μ on $\mathcal{C}^1 \times \mathcal{C}^2$, independent of (x_1, x_2), such that

$$\rho(y_1, y_2; x_1, x_2) = \sum_{c^1, c^2} \mu(c^1, c^2) \mathbf{1}\{c^1(x_1) = y_1\} \mathbf{1}\{c^2(x_2) = y_2\}. \tag{1}$$

The law μ may have arbitrary dependence. It allows correlated tastes, information, and randomization. The stable object is this law over complete choice functions, *not* the observed correlation of actions. Changes in a person's own menu may change both her action and the observed dependence between the two actions.

2 A binary example: one common overlap

For this example, impose no signalling and suppose decision maker 1 has two local covariate values, with two feasible actions at each. Let A_0, A_1 indicate whether she would choose a designated alternative at each. The unobserved common-population share

$$t = \Pr(A_0 = 1, A_1 = 1)$$

counts people who would choose that alternative in both situations. These counterfactual choices are not assumed to be jointly observed.

At a covariate j of decision maker 2, let B_j be her action. The observations identify, for each feasible b,

$$u_{jb} = \Pr(A_0 = 1, B_j = b), \quad v_{jb} = \Pr(A_1 = 1, B_j = b), \quad w_{jb} = \Pr(B_j = b).$$

The overlap within this group must lie between its unavoidable overlap and its available capacity. Summing over groups gives

$$\ell_j(\rho) = \sum_b \max\{0, u_{jb} + v_{jb} - w_{jb}\} \ \leq t \leq \ u_j(\rho) = \sum_b \min\{u_{jb}, v_{jb}\}. \tag{2}$$

For example, if 45 of a group of 50 people choose the designated action in each of two situations, at least $45 + 45 - 50 = 40$ must belong to both groups. No behavioral optimization is used: this is population accounting.

Base and recursive step. After the first covariate retain $L_1 = \ell_1$, $U_1 = u_1$. Then

$$\boxed{L_{n+1} = \max\{L_n, \ell_{n+1}\}, \qquad U_{n+1} = \min\{U_n, u_{n+1}\}.} \tag{3}$$

The same t must survive every observation. In this domain a simple excess score is

$$G_n^{\text{ov}} = L_n - U_n, \qquad F_n^{\text{ov}} = [L_n - U_n]_+ . \tag{4}$$

Under no signalling, $F_n^{\text{ov}} = 0$ is necessary and sufficient for separability for *any* finite number and cardinality of the second decision maker's covariates and action sets. Marginality fixes the other three cells of the common (A_0, A_1) distribution once t is fixed. The within-group intervals fill their full sum interval, so any common t constructs all required local tables.

New covariate	Its permitted range	Common range	$100F_n^{\text{ov}}$
First	40–50	40–50	0
Second	30–40	40	0
Third	0–30	None	10

Ranges are counts per 100 people. These arise from valid normalized marginal tables: let $w_{j0} = w_{j1} = 1/2$, both A marginals equal $1/2$, and (u_{j0}, v_{j0}) equal $(.45, .45)$, $(.45, .35)$, and $(.15, .35)$. The complementary $b = 1$ masses are $.5 - u_{j0}$ and $.5 - v_{j0}$. The first two observations allow one composition; the third conflicts with the first. This simplified score and the general construction below have the same zero set here; their numerical magnitudes need not coincide.

3 The general recursive formula

3.1 Start with explicit replica accounting

Enumerate X_1 by i and $X_2 = \{1, \ldots, m\}$ by j. Write $\alpha \in \mathcal{C}^1$ for a complete choice function of decision maker 1. Retain the common population $r(\alpha)$ and local shares $q_j(\alpha, b)$ satisfying

$$q_j(\alpha, b) \geq 0, \tag{5}$$

$$\sum_{\alpha:\alpha(i)=a} q_j(\alpha, b) = \rho(a, b; i, j), \tag{6}$$

$$\sum_b q_j(\alpha, b) = r(\alpha) \quad \text{for every } j, \alpha. \tag{7}$$

Here and below sums include only feasible actions. The replica test asks whether these equations have a nonnegative solution.

Collect all r and q coordinates in $z \in \mathbb{R}^N$. Encode each equality by its two opposite inequalities. The resulting finite source list has rows

$$a_e \cdot z \geq \ell_e(\rho), \qquad e \in E, \tag{8}$$

with domain-dependent integer coefficients and observed linear functionals ℓ_e. Specifically, nonnegativity has right side 0; the two signs of (6) have right sides $\rho(a, b; i, j)$ and $-\rho(a, b; i, j)$; and both signs of (7) have right side 0. This defines the initial rows without an unknown coefficient catalogue. Nonnegativity of r and its unit mass follow from (5)–(7) and observed normalization.

3.2 Two operations, repeated finitely many times

Represent a requirement by a pair (a, B), meaning $a \cdot z \geq B(\rho)$. The function B is a symbolic expression. Define Coalesce to replace all pairs with the same vector a by the single pair

$$\left(a, \max_{(a,B) \text{ present}} B\right).$$

Start with $\mathcal{S}_0 = \text{Coalesce}\{(a_e, \ell_e) : e \in E\}$. Fix an ordering of the coordinates $z_1, \ldots, z_N$. At stage k, partition the pairs in $\mathcal{S}_{k-1}$ according to the sign of a_k. The recursion is Fourier–Motzkin elimination with identical left sides coalesced. Budroni and Cabello [1] use variable elimination to derive Bell inequalities. In the present choice domain the steps are

$$\boxed{\begin{aligned}\mathcal{S}_k = \text{Coalesce}\Big(&\{(a, B) \in \mathcal{S}_{k-1} : a_k = 0\} \\ &\cup \{((-d_k)a + a_k d,\ (-d_k)B + a_k D) : \\ &\qquad (a, B), (d, D) \in \mathcal{S}_{k-1},\ a_k > 0 > d_k\}\Big).\end{aligned}} \tag{9}$$

Both multipliers in each new pair are positive, and its kth coefficient is zero. Keep *every* such pair and every already independent row. If only one sign occurs, there are no pairs of opposite signs; the zero-coefficient rows still survive.

Reporting convention. Use the unit source rows specified above, fix the finite index orders before seeing ρ, and use (9) without additional row rescaling. This fixes one score. Alternative orders or positive rescalings give equivalent tests, but need not give the same numerical score.

4 The finite observed H-representation

Every vector in $\mathcal{S}_N$ is zero. Define the nonnegative excess-coordination score by

$$\boxed{F_{\mathcal{D}}(\rho) = \max\Big(\{0\} \cup \{B(\rho) : (0, B) \in \mathcal{S}_N\}\Big).} \tag{10}$$

The convention also covers an empty terminal family. Equations (8), (9), and (10) are a finite recursive formula, not a search over data-dependent coefficient choices.

Theorem 1 (Recursive excess cross-covariate coordination)**.** *Fix any finite local-covariate domain $\mathcal{D}$ with nonempty feasible action sets and the reporting convention above. The recursion constructs a nonnegative, continuous, convex, piecewise-linear function $F_{\mathcal{D}}$ of the full observation array. There is a finite, nonempty, data-independent family $\mathcal{C}_{\mathcal{D}}$ of integer coefficient vectors, containing zero, such that*

$$F_{\mathcal{D}}(\rho) = \max_{c \in \mathcal{C}_{\mathcal{D}}} c \cdot \rho. \tag{11}$$

For every joint choice rule on this domain,

$$\boxed{\rho \text{ is separable} \iff F_{\mathcal{D}}(\rho) = 0 \iff c \cdot \rho \leq 0 \text{ for all } c \in \mathcal{C}_{\mathcal{D}}.} \tag{12}$$

Whenever $F_{\mathcal{D}}(\rho) > 0$, an attaining branch yields a violated observed linear inequality and a nonnegative source-row cancellation certificate. The family may contain redundant and non-facet-defining inequalities.

Proof. First, (1) implies the replica system by marginalization. Conversely, suppose that system is feasible. The common shares satisfy $r \geq 0$ and $\sum_\alpha r(\alpha) = 1$. For $r(\alpha) > 0$, define a distribution over the second decision maker's complete choice functions by

$$\Pr(c^2 = \beta \mid \alpha) = \prod_{j=1}^{m} \frac{q_j(\alpha, \beta(j))}{r(\alpha)}.$$

Conditional on α, this distribution has each prescribed j marginal. Mixing with r gives (1). When $r(\alpha) = 0$, all corresponding q_j are zero and that component receives zero mass. Conditional independence in this construction is a convenient existence proof, not a maintained assumption about behavior.

Second, coalescing is exact because all restrictions with one left side hold if and only if their maximum right side is satisfied. For fixed remaining coordinates and observations, a row with $a_k > 0$ gives a lower bound on z_k, and a row with $a_k < 0$ gives an upper bound. Every lower bound is at most every upper bound if and only if all pairs in (9) hold. Finiteness then gives a permissible z_k. If there are only lower bounds, only upper bounds, or no bounds, choose a sufficiently large value, a sufficiently small value, or any value, respectively. All restrictions independent of z_k remain. Thus each step is equivalent to existentially eliminating precisely that coordinate.

Induction over $k = 1, \ldots, N$ gives feasibility of the source system if and only if $B(\rho) \leq 0$ for every terminal pair. Equation (10) gives (12). Structural induction over maxima and positive sums gives the coefficient family and certificate described next. A finite maximum of linear forms is continuous, convex, and piecewise linear. All source coefficients and recursive multipliers are integers, so all branches have integer coefficients. □

5 Coefficients and violation certificates

For an observed linear leaf $\ell(\rho) = c \cdot \rho$, put $\mathcal{C}(\ell) = \{c\}$. Recursively define

$$\boxed{\begin{aligned} &\mathcal{C}(\max\{B, D\}) = \mathcal{C}(B) \cup \mathcal{C}(D), \\ &\quad \mathcal{C}(\lambda B + \eta D) = \{\lambda c + \eta d : c \in \mathcal{C}(B),\ d \in \mathcal{C}(D)\}, \qquad \lambda, \eta \geq 0. \end{aligned}} \tag{13}$$

Including the zero branch at the root gives $\mathcal{C}_{\mathcal{D}}$. The coefficient rule follows from

$$\lambda \max_c c \cdot \rho + \eta \max_d d \cdot \rho = \max_{c,d} (\lambda c + \eta d) \cdot \rho.$$

It is not necessary to expand this identity while evaluating the recursive expression. The fully expanded family may nevertheless be large; the theorem makes no polynomial-size claim.

5.1 A branch is a population-consistency argument

At a maximum, select an attaining requirement. At a positive sum, combine the selected requirements using their specified weights. Every resulting terminal branch has nonnegative source weights λ_e satisfying

$$\sum_e \lambda_e a_e = 0, \qquad c \cdot \rho = \sum_e \lambda_e \ell_e(\rho), \qquad -c \cdot \rho = \sum_e \lambda_e (a_e \cdot z - \ell_e(\rho)). \tag{14}$$

The equality rows have zero residual whenever the accounting equalities hold. The nonnegativity rows have residuals equal to the corresponding replica coordinates. Consequently, if a nonnegative candidate table satisfying the accounting equalities existed, the right side would be a nonnegative weighted sum of genuine population shares. The hidden population coefficients have canceled. If $c \cdot \rho > 0$, that accounting identity would require this nonnegative sum to be negative.

This is a precise sense in which the inequalities reveal *excess coordination*: observations impose requirements that cannot fit in one stable population. A maximum picks the strongest requirement on the same quantity. A positive sum balances requirements about a hidden quantity so it can be removed. The combined groups need not be disjoint; the operation is weighted accounting, not an unsupported addition of head counts.

5.2 What the score measures

For the fixed construction, $F_{\mathcal{D}}$ is a nonnegative *excess-coordination index*. Zero means that a separable explanation exists; it does not mean little dependence or absence of real-world interaction. Perfect agreement can be separable. A positive value supplies an explicit reason why separate responses with stable shared influences cannot explain the whole array.

The magnitude is construction-specific. Rescaling inequalities or changing the elimination convention can change it. It is not a canonical distance, a Pearson correlation, a social multiplier, or an identified causal peer-effect magnitude. A comparison of numerical scores must state the same domain, coding, and normalization convention. In finite samples, a positive plug-in score by itself is not a significance test.

Individual generated rows are valid bounds, but their zero bounds need not all be attained on normalized separable data. Some are loose or redundant. A complete H-representation is not the same object as a minimal list of sharp facets.

6 Adding covariates

Theorem 1 eliminates hidden coordinates. There is also an economic induction that *adds covariate values*. Keep decision maker 1's domain fixed. Let $\mathcal{P}_j(\rho)$ be the set of common populations r for which the local shares q_j satisfy (5)–(7) at covariate j. Define

$$\boxed{K_1(\rho) = \mathcal{P}_1(\rho), \qquad K_{n+1}(\rho) = K_n(\rho) \cap \mathcal{P}_{n+1}(\rho).} \tag{15}$$

By induction, K_n is exactly the population compositions that can explain every processed covariate. It is nonempty if and only if the restricted observation array is separable. With only one covariate for decision maker 2, normalized probability data satisfying no signalling always admit such an explanation: within each action group, couple decision maker 1's specified single-covariate choices independently. This establishes a genuine base case without assuming a preferred extension.

The recursive state is K_n, not merely the number F_n. Two intervals may have the same gap but be located differently; only one may overlap a new requirement. In higher dimension, separate ranges

for each population share also lose joint trade-offs. Retain an exact description of the whole ledger and apply the symbolic elimination when an observed score or certificate is needed.

If a fixed complete list of accounting directions h describes each local ledger by $h \cdot r \geq B_{h,j}(\rho)$, the same induction has the compact form

$$R_h^{(1)} = B_{h,1}, \qquad R_h^{(n+1)} = \max\{R_h^{(n)}, B_{h,n+1}\}.$$

This update applies whenever the direction list is complete for the chosen domain. The direct construction in (9) works from the original accounting rows.

The sets K_n shrink, so adding data cannot repair an already impossible common-population explanation. This does *not* assert numerical monotonicity of generic scores rebuilt with different elimination orders or scales on growing domains. With a fixed catalogue and fixed positive weights the demand-envelope scores are monotone; that is a separate, stronger statement about a specified representation.

6.1 Stable influences, not stable observed correlation

The distinction persists even with rational choice. Let decision maker 1 face either $\{x, y\}$ or $\{x, z\}$, and decision maker 2 face $\{u, v\}$. Two equally likely joint types have preferences and choices

Type	Decision maker 1's ranking	Decision maker 2's choice
First	$z \succ x \succ y$	u
Second	$y \succ x \succ z$	v

Score choosing x as $+1$ and the other available option as -1; score u as $+1$ and v as -1. Correlation changes from $+1$ at $\{x, y\}$ to -1 at $\{x, z\}$. The same distribution over types explains both, with uniform scored marginals. Thus neither a change in correlation nor high correlation alone is excess coordination.

7 Adding a menu: finite coefficients for a fixed score

Fix one decision maker's original menus and choose one reference action in each. For an old choice function α, let $f(\alpha) \in \{0,1\}^d$ list the nonreference action indicators, where d is the sum of the original menu sizes minus one per menu. Add a new menu with actions indexed by $a = 0, \ldots, k-1$. In a fixed menu/outcome cell of the other decision maker, write p_a for the mass choosing a from the new menu and s for the old-menu nonreference masses. Assume $p, s \geq 0$, and that the nonreference components for each old menu sum to at most $\sum_a p_a$.

For a fixed real score h on the extended choice functions, define

$$\phi_h(s,p) = \min_{q \geq 0} \left\{ \sum_{\alpha,a} h(\alpha,a) q_{\alpha a} : \sum_{\alpha} q_{\alpha a} = p_a, \quad \sum_{\alpha,a} f(\alpha) q_{\alpha a} = s \right\}.$$

The assumed margins suffice for feasibility by independent coupling, including the zero-mass case. The total mass constraint makes the feasible set compact. Let

$$\kappa_a(\theta) = \min_{\alpha} \{h(\alpha,a) - \theta \cdot f(\alpha)\}.$$

Finite linear programming duality gives

$$\phi_h(s,p) = \max_{\theta \in \mathbb{R}^d} \left\{ \theta \cdot s + \sum_a p_a \kappa_a(\theta) \right\}. \tag{16}$$

Indeed, the dual has constraints $\theta \cdot f(\alpha) + \kappa_a \leq h(\alpha, a)$. For fixed θ, the objective is maximized by setting each κ_a at its upper bound, since its weight is $p_a \geq 0$.

The finite coefficient formula is obtained by forming all within-category equations

$$\theta \cdot [f(\alpha) - f(\alpha')] = h(\alpha, a) - h(\alpha', a). \tag{17}$$

Select any d equations with independent normals and solve their square system, $\theta = A^{-1}b$. Let V_h be the finite set of resulting candidates, and use the singleton empty vector if $d = 0$. In (16) one may replace $\mathbb{R}^d$ by V_h. To see this, the dual polyhedron contains no lines: the all-reference old plan forces all categorical components of a line direction to vanish, and single-coordinate deviations force the remaining components to vanish. A finite optimum is attained at a vertex. At such a vertex, choose one active equation per category, and subtract it from the other active equations in that category. The resulting within-category equations must span the d remaining directions. Consequently an optimal slope belongs to V_h. Every extra slope in V_h, with κ computed as above, gives a feasible dual objective.

There is also a direct recursion in the original variables. Write s_a for the old-menu nonreference shares within category a. Split $s = \sum_a s_a$ into feasible shares for masses p_a, then minimize $\sum_a \phi^{\text{old}}_{h(\cdot,a)}(s_a, p_a)$, where each old problem minimizes over complete old choice functions with the indicated mass and shares. Splitting the columns of q proves one direction and concatenating attaining columns proves the other. For three binary menus followed by a ternary menu, $d = 3$. Equations (16)–(17) evaluate a given score on all 24 choice functions using three slope coordinates. A separability test also needs sufficient scores to compare across cells.

8 A parametric test for the full three-alternative menu domain

Each decision maker now faces three two-action menus and one three-action menu, so the domain is $(2, 2, 2, 3)^2$. Maintain no signalling in this section. Write X_i for the selected-action indicator at binary menu $i = 1, 2, 3$ of decision maker 1, T for her ternary action, and B_j for decision maker 2's choice at menu $j = 0, 1, 2, 3$. The observed masses

$$p_{jab} = \Pr(T = a, B_j = b), \qquad r_{ijb} = \Pr(X_i = 1, B_j = b)$$

give the following compact set P of candidate allocations for the unobserved category shares m_{ijab}:

$$0 \leq m_{ijab} \leq p_{jab}, \qquad \sum_{a=0}^{2} m_{ijab} = r_{ijb}. \tag{18}$$

There are 27 capped triples, one per (i, j, b). The observed margins guarantee that P is nonempty. For fixed a, set the two cells of each binary layer to m_{ijab} and $p_{jab} - m_{ijab}$. They sum in every menu pair to the same category mass $\Pr(T = a)$ by no signalling. Apply the homogeneous H rows of Theorem 1 to each layer, with signs written as $\ell_l(m; p) \geq 0$. Positive-mass layers can be normalized before applying the theorem. Zero-mass layers have all cells zero.

For a separable rule, its joint plan law supplies an allocation in P with all the layer inequalities. Conversely, given such an allocation, a separable joint plan law in each layer can be combined with its category mass and with the ternary action fixed to a. The combined law reproduces the observations because the binary cells sum to r_{ijb} and their complements, while the ternary cells are p_{jab}. For the projection it is convenient also to include the balance equations

$$E_{iaj}(m) = \sum_b m_{ijab} - \sum_b m_{i0ab} = 0 \quad (i = 1, 2, 3,\ a = 0, 1,\ j = 1, 2, 3).$$

The layer inequalities already imply these balances. The omitted category-2 balances follow from the cell totals and no signalling. In this chart, the 81 allocations have 36 coordinates after the 27 cell sums and 18 balances, before imposing the box bounds.

For any nonnegative weights w_l and arbitrary real balance coefficients η_{iaj}, write $\sum_l w_l \ell_l(m;p) + \sum_{i,a,j} \eta_{iaj} E_{iaj}(m)$ as

$$K_w(p) + \sum_{i,j,b,a} c_{ijba}(w,\eta) m_{ijab}.$$

For each triple let $p_{jb} = (p_{j0b}, p_{j1b}, p_{j2b})$ and write c_{ijb} for its vector of coefficients. Define

$$\Phi(c;p,r) = \min_{t\in\{0,1,2\}} \left\{ c_t r + \sum_{a=0}^{2} p_a (c_a - c_t)_+ \right\}.$$

This is the maximum of $c \cdot m$ under the corresponding triple in (18). Each threshold gives an upper bound by writing $c \cdot m = c_t r + \sum_a (c_a - c_t) m_a$. Fill categories in decreasing price order and choose the stopping price as the threshold to attain the bound, with the highest or lowest price at the two endpoints.

The complete observed test is

$$K_w(p) + \sum_{i,j,b} \Phi(c_{ijb}(w,\eta); p_{jb}, r_{ijb}) \geq 0 \qquad \text{for all } w \geq 0,\ \eta \in \mathbb{R}^{18}. \tag{19}$$

A feasible layer allocation makes each weighted score nonnegative and the balance term zero, so its maximum over P is nonnegative. Conversely, if no such allocation exists, the compact convex image $\{((\ell_l(m;p))_l, E(m)) : m \in P\}$ is disjoint from the nonnegative orthant times $\{0\}$. Strict separation gives $w \geq 0$ and unrestricted η with a strictly negative maximum over P. The local support identity computes that maximum exactly, contradicting (19). For fixed parameters, choosing a threshold independently in each triple yields observed linear rows, all of which must hold. The parameters range over a complete family. A small sufficient parameter list and a compact facet catalogue for this domain remain open.

9 Scope and verification

The finite elimination theorem for unrestricted choice has a Lean proof using the opposite replica orientation, with a common distribution over decision maker 2's complete plans. Lean also checks coefficient semantics for finite maximum and nonnegative-sum expressions, common-population induction, and the generic projection through capped triples. The proofs above specify the economic substitution and the grouped score directly. Verification of the executable compiler and a compact catalogue are separate questions.

References

[1] C. Budroni and A. Cabello (2012). Bell inequalities from variable-elimination methods. *Journal of Physics A: Mathematical and Theoretical* 45, 385304.

[2] N. Kashaev, M. Plávala, and V. H. Aguiar (2026). Entangled vs. Separable Choice. Working paper, arXiv:2403.09045v4.